

IC 130: Applied Chemistry for Engineers

Course Code	IC 130
Course Name	Applied Chemistry for Engineers
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum; 29th BoA (Mar 2019), in the B.Tech-M.Tech IDD Bioengineering curriculum; 22nd Senate (Jun 2019)

Course Contents

Part I – Classical mechanics

- **Spectroscopy**- Introduction and classification, Fundamental principles, Instrumentation and applications of Ultra Violet-Visible Spectroscopy, Infra-Red Spectroscopy, Raman Spectroscopy and Nuclear Magnetic Resonance Spectroscopy [12 Lectures]
- **Polymer Chemistry**- Introduction, Polymerisation, Properties, Polymer processing, Industrial polymers, conducting polymers [8 Lectures]
- **Fuels and Combustion**- Properties of fuels, Calorific value, Petroleum and petrochemicals, biofuels [6 Lectures]
- **Electrochemistry**- Applications of electrochemistry at the interface of science and technology, Batteries, Fuel cells, Biomedical devices, Corrosion and its control [10 Lectures]
- **Lubricants**- Mechanism of lubrication, Types, Properties and selection of lubricants [6 Lectures]

Text Books

1. H.D. Gesser, **Applied Chemistry - A Textbook for Engineers and Technologists**, Springer
2. Wiley India Editorial Team, **Engineering Chemistry**, Wiley India Pvt. Ltd., 2011.
3. Shashi Chawla, **Engineering Chemistry**

References

1. J. M. Hollas, **Modern Spectroscopy**, Wiley India Pvt. Ltd.
2. Colin Banwell and Elaine McCash, M A Ruderman, **Fundamentals of molecular spectroscopy**, Tata McGraw Hill Education Pvt. Ltd.
3. Fred W. Billmeyer, **Text Book Polymer Science**, Wiley India Pvt. Ltd.